

Mycelic

Loss accounting, gap decomposition and vNext progress

QUESTION

How should thousands of private user-level agents abstract, route, combine, question, verify and propagate what they know upward, so that an enterprise kernel discovers what no individual could?

ENTERPRISE SIZES MEASURED

— users

CONTENTS

9 sections · 33 tables · 0 figures

GENERATED

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Every table and figure in this document is computed from the raw per-run records in `research/mycelic/artifacts/` at generation time. No number is transcribed by hand. Section 5 states exactly which results were measured on real models and which were simulated.

What this is. The benchmark report says the hierarchy finds 36% of the hidden patterns at 50,000 users and the strongest centralised system finds 70%. This document says *where the other patterns went*. Every discoverable gold pattern is traced through the pipeline and the first stage at which it is lost is recorded, for the hierarchy (`H_mycelic_full`), its centralised twin (`B4_central_triage`), the perfect-retrieval oracle (`Y_oracle_retrieval`) and the strongest centralised baseline (`A2_chunked_ctx`), on the same worlds. Then three paired diagnostics on identical worlds decompose the gap into the stages responsible.

Nothing here is a design. It is the measurement the next design is built on. Every number is computed from

`research/mycelic/artifacts/loss_funnel.jsonl` and `quick_diag*.jsonl` at generation time. 5 seeds at 10,000 users, 2 at 50,000.

1. The finding in one paragraph

The handoff pack's leading hypothesis was that the sketch channel is too poor to seed good hypotheses. It is not: at 10,000 users the sketch triage sees 98.5% of gold anchors. The pattern dies later, at two places. First, the **question budget**: the calibrated budget asks only 63.5% of gold anchors at 10k and 43% at 50k, and that is the single largest first-loss stage at both scales (47% and 51% of the gap to A2). Second, the **register**: at 10k, 12% of all patterns are matched at confidence ≥ 0.5 and then cut, because the hierarchy rates spurious candidates as highly as genuine ones and the register holds one entry per entity. A2 loses nothing at the register. Asking every triage candidate recovers evidence for 98.5% of patterns and the kernel forms a correct confident candidate for ~80% — more than A2 — and then buries them. The calibrated small budget was the right choice *given the ranker*; the ranker is the fault. At 50,000 users the sketch additionally loses 18% of patterns (mostly rare: a single-witness facet cannot set a site bit under the support threshold), so scale adds a third, smaller stage.

2. Stage-by-stage survival

Stages are ordered by pipeline position. They are not strictly nested — a pattern can be visible to the sketch without having been extracted — so `first stage lost` records the first stage in order that failed. The flat systems have no sketch, triage, question or descent stage.

10,000 users (5 seeds)

Stage survival, all patterns

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (≥ 2 facets, right entity+predicate)	—	0.960 [0.93, 0.99]	0.960 [0.93, 0.99]	0.960 [0.93, 0.99]
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	—	—	0.985 [0.97, 0.99]	—
on the triage candidate list	—	—	0.985 [0.97, 0.99]	—
inside the question budget	—	—	0.635 [0.58, 0.70]	—
descent reached a facet holder	—	—	0.665 [0.61, 0.73]	—
≥ 2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.970 [0.95, 0.99]	0.690 [0.64, 0.74]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.885 [0.84, 0.92]	0.870 [0.82, 0.92]	0.595 [0.54, 0.65]	0.865 [0.85, 0.89]
matched primary rule at ANY confidence	0.855 [0.80, 0.89]	0.770 [0.74, 0.81]	0.540 [0.45, 0.63]	0.770 [0.72, 0.82]
matched with confidence ≥ 0.5	0.810 [0.74, 0.86]	0.760 [0.73, 0.79]	0.515 [0.44, 0.61]	0.745 [0.70, 0.79]
survived the register cut (= reported)	0.685 [0.62, 0.74]	0.370 [0.32, 0.42]	0.375 [0.32, 0.43]	0.240 [0.19, 0.28]

patterns counted: A2_chunked_ctx: 200, B4_central_triage: 200, H_mycelic_full: 200, Y_oracle_retrieval: 200

Stage survival, rare-signal patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (≥ 2 facets, right entity+predicate)	—	0.910 [0.83, 0.99]	0.910 [0.83, 0.99]	0.910 [0.83, 0.99]
visible to sketch triage (≥ 2 foreign sites, ≥ 2 regions, span ≥ 2)	—	—	0.977 [0.95, 1.00]	—

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
on the triage candidate list	—	—	0.977 [0.95, 1.00]	—
inside the question budget	—	—	0.446 [0.36, 0.54]	—
descent reached a facet holder	—	—	0.463 [0.41, 0.53]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.955 [0.92, 0.99]	0.500 [0.42, 0.58]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.875 [0.82, 0.92]	0.801 [0.76, 0.86]	0.416 [0.30, 0.54]	0.853 [0.82, 0.89]
matched primary rule at ANY confidence	0.851 [0.75, 0.92]	0.703 [0.65, 0.77]	0.360 [0.26, 0.46]	0.772 [0.69, 0.85]
matched with confidence >= 0.5	0.773 [0.62, 0.90]	0.703 [0.65, 0.77]	0.341 [0.25, 0.45]	0.737 [0.67, 0.80]
survived the register cut (= reported)	0.653 [0.55, 0.75]	0.258 [0.19, 0.33]	0.215 [0.16, 0.27]	0.143 [0.10, 0.18]

patterns counted: A2_chunked_ctx: 78, B4_central_triage: 78, H_mycelic_full: 78, Y_oracle_retrieval: 78

Stage survival, common patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.993 [0.98, 1.00]	—
on the triage candidate list	—	—	0.993 [0.98, 1.00]	—
inside the question budget	—	—	0.771 [0.66, 0.85]	—
descent reached a facet holder	—	—	0.798 [0.72, 0.88]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.986 [0.96, 1.00]	0.821 [0.76, 0.88]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.893 [0.86, 0.93]	0.916 [0.86, 0.98]	0.721 [0.69, 0.76]	0.870 [0.84, 0.90]
matched primary rule at ANY confidence	0.860 [0.84, 0.89]	0.818 [0.79, 0.85]	0.660 [0.56, 0.75]	0.761 [0.72, 0.81]
matched with confidence >= 0.5	0.845 [0.83, 0.86]	0.804 [0.77, 0.84]	0.630 [0.54, 0.72]	0.746 [0.70, 0.79]
survived the register cut (= reported)	0.716 [0.67, 0.76]	0.450 [0.34, 0.55]	0.484 [0.43, 0.55]	0.307 [0.25, 0.35]

patterns counted: A2_chunked_ctx: 122, B4_central_triage: 122, H_mycelic_full: 122, Y_oracle_retrieval: 122

Where the hierarchy loses each pattern (terminal stage)

stage the pattern died at	all	rare	common
extracted (>=2 facets, right entity+predicate)	0 (0%)	0	0
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	3 (2%)	2	1
on the triage candidate list	0 (0%)	0	0
inside the question budget	58 (29%)	37	21
descent reached a facet holder	0 (0%)	0	0
>=2 gold links in kernel pool (= evidence coverage)	0 (0%)	0	0
candidate formed on right entity + chain	17 (8%)	7	10
matched primary rule at ANY confidence	14 (7%)	5	9
matched, but confidence < 0.5	5 (2%)	1	4
cut from the register (matched at >= 0.5, out-ranked)	28 (14%)	10	18
reported	75 (38%)	16	59
total	200	78	122

Why the sketch could not see the invisible ones

why the sketch could not see it	all	rare	common	median witnesses
causal span < 2 across foreign sites	3	2	1	7

Discovery by number of witnesses (hierarchy)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	12	1.00	0.75	0.25
5–7	52	0.96	0.46	0.21
8–11	16	1.00	0.44	0.12
12–16	13	1.00	0.62	0.15
17–25	65	0.98	0.78	0.45
26–∞	42	1.00	0.93	0.67

Discovery by number of witnesses (A2_chunked_ctx)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	12	—	1.00	0.67
5–7	52	—	1.00	0.56
8–11	16	—	1.00	0.81
12–16	13	—	1.00	0.77
17–25	65	—	1.00	0.74
26–∞	42	—	1.00	0.69

Where the reported patterns sit in the register

architecture	reported	median rank (0-based, confidence-sorted register)	within top 40	within top 100	median confidence
A2_chunked_ctx	137	197	19	37	0.85
B4_central_triage	74	263	9	16	0.88
H_mycelic_full	75	255	8	18	0.89
Y_oracle_retrieval	48	279	3	9	0.91

Decomposing the gap to the strongest centralised system

Paired on 200 (seed, pattern) pairs at 10,000 users: A2_chunked_ctx reports 137 (68.5%), H_mycelic_full reports 75 (37.5%). A2_chunked_ctx finds 77 that H_mycelic_full misses; H_mycelic_full finds 15 that A2_chunked_ctx misses.

Of the 77 patterns A2_chunked_ctx reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage it died at	patterns	share of the gap	of which rare
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	1	1%	1
inside the question budget	36	47%	24
candidate formed on right entity + chain	7	9%	3
matched primary rule at ANY confidence	7	9%	3
matched, but confidence < 0.5	3	4%	0
cut from the register (matched at >= 0.5, out-ranked)	23	30%	7

Decomposing the gap to the centralised twin (same algorithm)

Paired on 200 (seed, pattern) pairs at 10,000 users: B4_central_triage reports 74 (37.0%), H_mycelic_full reports 75 (37.5%). B4_central_triage finds 34 that H_mycelic_full misses; H_mycelic_full finds 35 that B4_central_triage misses.

Of the 34 patterns **B4_central_triage** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
inside the question budget	16	47%	6
candidate formed on right entity + chain	4	12%	1
matched primary rule at ANY confidence	3	9%	1
matched, but confidence < 0.5	3	9%	1
cut from the register (matched at >= 0.5, out-ranked)	8	24%	1

Decomposing the gap to the perfect-retrieval oracle

Paired on 200 (seed, pattern) pairs at 10,000 users: **Y_oracle_retrieval** reports 48 (24.0%), **H_mycelic_full** reports 75 (37.5%). **Y_oracle_retrieval** finds 16 that **H_mycelic_full** misses; **H_mycelic_full** finds 43 that **Y_oracle_retrieval** misses.

Of the 16 patterns **Y_oracle_retrieval** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
inside the question budget	10	62%	6
candidate formed on right entity + chain	1	6%	0
matched primary rule at ANY confidence	2	12%	1
matched, but confidence < 0.5	1	6%	0
cut from the register (matched at >= 0.5, out-ranked)	2	12%	0

50,000 users (2 seeds)

Stage survival, all patterns

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	0.940 [0.93, 0.95]	0.940 [0.93, 0.95]	0.930 [0.93, 0.93]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.770 [0.74, 0.80]	—
on the triage candidate list	—	—	0.770 [0.74, 0.80]	—
inside the question budget	—	—	0.405 [0.36, 0.45]	—
descent reached a facet holder	—	—	0.400 [0.35, 0.45]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.490 [0.47, 0.51]	0.430 [0.38, 0.48]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.870 [0.87, 0.87]	0.375 [0.34, 0.41]	0.375 [0.34, 0.41]	0.870 [0.87, 0.87]
matched primary rule at ANY confidence	0.840 [0.84, 0.84]	0.325 [0.31, 0.34]	0.350 [0.31, 0.39]	0.790 [0.79, 0.79]
matched with confidence >= 0.5	0.790 [0.79, 0.79]	0.325 [0.31, 0.34]	0.350 [0.31, 0.39]	0.780 [0.78, 0.78]
survived the register cut (= reported)	0.690 [0.69, 0.69]	0.325 [0.31, 0.34]	0.350 [0.31, 0.39]	0.160 [0.16, 0.16]

patterns counted: A2_chunked_ctx: 100, B4_central_triage: 200, H_mycelic_full: 200, Y_oracle_retrieval: 100

Stage survival, rare-signal patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	0.858 [0.84, 0.87]	0.858 [0.84, 0.87]	0.844 [0.84, 0.84]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.551 [0.44, 0.67]	—
on the triage candidate list	—	—	0.551 [0.44, 0.67]	—

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
inside the question budget	—	—	0.282 [0.23, 0.33]	—
descent reached a facet holder	—	—	0.232 [0.15, 0.31]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.212 [0.18, 0.24]	0.293 [0.23, 0.36]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.889 [0.89, 0.89]	0.142 [0.13, 0.16]	0.234 [0.18, 0.29]	0.822 [0.82, 0.82]
matched primary rule at ANY confidence	0.867 [0.87, 0.87]	0.129 [0.10, 0.16]	0.210 [0.15, 0.27]	0.778 [0.78, 0.78]
matched with confidence >= 0.5	0.778 [0.78, 0.78]	0.129 [0.10, 0.16]	0.210 [0.15, 0.27]	0.778 [0.78, 0.78]
survived the register cut (= reported)	0.622 [0.62, 0.62]	0.129 [0.10, 0.16]	0.210 [0.15, 0.27]	0.111 [0.11, 0.11]

patterns counted: A2_chunked_ctx: 45, B4_central_triage: 84, H_mycelic_full: 84, Y_oracle_retrieval: 45

Stage survival, common patterns only

stage	A2_chunked_ctx	B4_central_triage	H_mycelic_full	Y_oracle_retrieval
extracted (>=2 facets, right entity+predicate)	—	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]	1.000 [1.00, 1.00]
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	—	—	0.922 [0.91, 0.93]	—
on the triage candidate list	—	—	0.922 [0.91, 0.93]	—
inside the question budget	—	—	0.494 [0.44, 0.55]	—
descent reached a facet holder	—	—	0.520 [0.48, 0.56]	—
>=2 gold links in kernel pool (= evidence coverage)	1.000 [1.00, 1.00]	0.688 [0.65, 0.72]	0.529 [0.48, 0.58]	1.000 [1.00, 1.00]
candidate formed on right entity + chain	0.855 [0.85, 0.85]	0.541 [0.49, 0.59]	0.476 [0.44, 0.51]	0.909 [0.91, 0.91]
matched primary rule at ANY confidence	0.818 [0.82, 0.82]	0.464 [0.44, 0.49]	0.450 [0.41, 0.49]	0.800 [0.80, 0.80]
matched with confidence >= 0.5	0.800 [0.80, 0.80]	0.464 [0.44, 0.49]	0.450 [0.41, 0.49]	0.782 [0.78, 0.78]
survived the register cut (= reported)	0.745 [0.75, 0.75]	0.464 [0.44, 0.49]	0.450 [0.41, 0.49]	0.200 [0.20, 0.20]

patterns counted: A2_chunked_ctx: 55, B4_central_triage: 116, H_mycelic_full: 116, Y_oracle_retrieval: 55

Where the hierarchy loses each pattern (terminal stage)

stage the pattern died at	all	rare	common
extracted (>=2 facets, right entity+predicate)	5 (2%)	5	0
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	40 (20%)	31	9
on the triage candidate list	0 (0%)	0	0
inside the question budget	69 (34%)	23	46
descent reached a facet holder	0 (0%)	0	0
>=2 gold links in kernel pool (= evidence coverage)	0 (0%)	0	0
candidate formed on right entity + chain	11 (6%)	5	6
matched primary rule at ANY confidence	5 (2%)	2	3
matched, but confidence < 0.5	0 (0%)	0	0
cut from the register (matched at >= 0.5, out-ranked)	0 (0%)	0	0
reported	70 (35%)	18	52
total	200	84	116

Why the sketch could not see the invisible ones

why the sketch could not see it	all	rare	common	median witnesses
fewer than 2 foreign sites	9	6	3	6

why the sketch could not see it	all	rare	common	median witnesses
foreign sites in fewer than 2 regions	6	6	0	6
causal span < 2 across foreign sites	31	25	6	7

Discovery by number of witnesses (hierarchy)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	11	0.55	0.55	0.36
5–7	57	0.56	0.26	0.18
8–11	16	0.56	0.25	0.25
12–16	17	0.82	0.24	0.18
17–25	61	0.92	0.49	0.41
26–∞	38	0.97	0.71	0.63

Discovery by number of witnesses (A2_chunked_ctx)

witnesses (records)	n	visible to sketch	in pool	reported
0–4	7	—	1.00	0.57
5–7	31	—	1.00	0.58
8–11	7	—	1.00	0.86
12–16	7	—	1.00	0.71
17–25	33	—	1.00	0.73
26–∞	15	—	1.00	0.80

Where the reported patterns sit in the register

architecture	reported	median rank (0-based, confidence-sorted register)	within top 40	within top 100	median confidence
A2_chunked_ctx	69	895	3	11	0.84
B4_central_triage	65	670	2	5	0.84
H_mycelic_full	70	877	1	3	0.88
Y_oracle_retrieval	16	850	0	1	0.95

Decomposing the gap to the strongest centralised system

Paired on 100 (seed, pattern) pairs at 50,000 users: A2_chunked_ctx reports 69 (69.0%), H_mycelic_full reports 39 (39.0%). A2_chunked_ctx finds 37 that H_mycelic_full misses; H_mycelic_full finds 7 that A2_chunked_ctx misses.

Of the 37 patterns A2_chunked_ctx reports and H_mycelic_full does not, the first stage H_mycelic_full lost them at:

stage it died at	patterns	share of the gap	of which rare
extracted (>=2 facets, right entity+predicate)	1	3%	1
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	9	24%	6
inside the question budget	21	57%	9
candidate formed on right entity + chain	4	11%	2
matched primary rule at ANY confidence	2	5%	1

Decomposing the gap to the centralised twin (same algorithm)

Paired on 200 (seed, pattern) pairs at 50,000 users: B4_central_triage reports 65 (32.5%), H_mycelic_full reports 70 (35.0%). B4_central_triage finds 21 that H_mycelic_full misses; H_mycelic_full finds 26 that B4_central_triage misses.

Of the 21 patterns **B4_central_triage** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
inside the question budget	17	81%	0
candidate formed on right entity + chain	2	10%	0
matched primary rule at ANY confidence	2	10%	1

Decomposing the gap to the perfect-retrieval oracle

Paired on 100 (seed, pattern) pairs at 50,000 users: **Y_oracle_retrieval** reports 16 (16.0%), **H_mycelic_full** reports 39 (39.0%). **Y_oracle_retrieval** finds 7 that **H_mycelic_full** misses; **H_mycelic_full** finds 30 that **Y_oracle_retrieval** misses.

Of the 7 patterns **Y_oracle_retrieval** reports and **H_mycelic_full** does not, the first stage **H_mycelic_full** lost them at:

stage it died at	patterns	share of the gap	of which rare
visible to sketch triage (>=2 foreign sites, >=2 regions, span>=2)	3	43%	2
inside the question budget	3	43%	0
candidate formed on right entity + chain	1	14%	0

3. Paired decomposition on identical worlds

Each row is the calibrated hierarchy with one setting changed, run on the same (scale, seed) worlds as the base. **An unbounded register is a benchmark change, not a design**; it appears here only to measure how much discovery is being lost to ranking rather than to retrieval or judgement. Arrows mark a 95% bootstrap interval entirely on one side of zero.

10,000 users

variant	found	evidence cov.	rare recall	AP	FDR	compute	calls
H_mycelic_full (calibrated)	0.375	0.690	0.215	0.0227	0.975	1.09e+06	9.10e+04
ask every triage candidate (question_frac = 1.0) (5 seeds)	0.300	0.985 ↑	0.271	0.0113 ↓	0.980	2.14e+06 ↑	2.21e+05 ↑
unbounded register (diagnostic only) (5 seeds)	0.540 ↑	0.690	0.341 ↑	0.0261 ↑	0.985 ↑	1.16e+06 ↑	9.10e+04
both (5 seeds)	0.795 ↑	0.985 ↑	0.697 ↑	0.0193	0.989 ↑	2.36e+06 ↑	2.21e+05 ↑
full budget, home-site evidence allowed (5 seeds)	0.360	0.985 ↑	0.274	0.0209	0.976	2.76e+06 ↑	2.56e+05 ↑

50,000 users

variant	found	evidence cov.	rare recall	AP	FDR	compute	calls
H_mycelic_full (calibrated)	0.363	0.450	0.235	0.0081	0.987	2.97e+06	2.89e+05
ask every triage candidate (question_frac = 1.0) (3 seeds)	0.407 ↑	0.787 ↑	0.248	0.0085	0.986 ↓	6.27e+06 ↑	6.99e+05 ↑
unbounded register (diagnostic only) (3 seeds)	0.363	0.450	0.235	0.0081	0.987	2.98e+06 ↑	2.89e+05
both (3 seeds)	0.643 ↑	0.787 ↑	0.417 ↑	0.0112 ↑	0.993 ↑	6.83e+06 ↑	6.99e+05 ↑

Reading it. Asking every triage candidate raises evidence coverage from 0.69 to 0.985 at 10k (0.45 to 0.79 at 50k) and *lowers* discovery at 10k: the recovered evidence produces candidates the register cannot hold. Remove the register pressure and the same runs report 0.795 at 10k — above A2's 0.685 — and 0.643 at 50k against A2's 0.703, at two-thirds of A2's compute. Allowing home-site evidence back in does not help, so the flood is not the foreign-only filter. The residual at 50k is the sketch loss, which no question budget can recover because the anchor never reaches the triage list.

4. Why the hierarchy's ranker is worse than A2's on the same operator

Both systems rank with the same confidence logistic. Its main terms — a causal span, several regions, links dominated by different branches — are exactly the properties the triage *pre-selects* anchors for. Among the candidates the hierarchy's kernel actually has to order, those terms carry almost no information, because every questioned anchor already has them. A2's pool holds every anchor, including thousands with no cross-organisational structure at all, for which the same terms are decisive. The hierarchy's discrimination problem is a conditioning problem, not an operator problem.

The consequence is testable: features the triage did **not** condition on — per-link independent support, inter-link lags, conflict volume, dispersion, whether the candidate came from a question — should separate genuine from spurious candidates among the triage survivors. Section 6 measures that.

5. Two smaller losses, measured

Extraction (10% at 50k, 24% of rare patterns). This is stochastic recall, not a capability ceiling: a fresh edge-tier re-read of the same records recovers 80–91% of the patterns lost there, a kernel-tier *local* re-read 100%. A user agent asked a targeted question about an entity can re-read its own notes about it — the entity name is in every record's surface text — so the loss is recoverable without moving any text.

Sketch (18% at 50k, 39 of 54 rare). 43 of the 54 fail on "causal span < 2 across foreign sites": a facet with one witness at a site sets no bit under `sketch_min_support = 2`. Lowering the threshold lets benign single mentions set bits and lengthens the triage list, which only pays if the ranker can absorb the noise — which is the same lever as section 4.

6. How separable are genuine candidates, on kernel-side features?

Every candidate the synthesis produced (before the register cut) was dumped with its confidence-logistic inputs and labelled against gold. One shared logistic was fitted on the pooled candidates of all four systems on the calibration seeds (500–502) and evaluated on the evaluation seeds (0–4) by re-ranking each run's candidates under its own register cap. No feature reads raw text; the privacy accounting is unchanged.

Dump default budget — 10,074 training candidates (seeds 500–502), 16,758 evaluation candidates (seeds 0–4), positive rate 2.4%. Pooled AUC gold-vs-spurious: hand-set confidence 0.609, learned 0.785.

architecture	gold patterns	with any candidate	found (current ranker)	found (learned rank, same #kept)	found (learned, p≥0.5)	AUC current	AUC learned
A2_chunked_ctx	200	137	0.685	0.685	0.495	0.630	0.736
B4_central_triage	200	74	0.370	0.370	0.220	0.562	0.778
H_mycelic_full	200	108	0.375	0.450	0.330	0.692	0.800
Y_oracle_retrieval	200	48	0.240	0.240	0.150	0.521	0.728

Largest standardised weights: `dispersion` -1.29, `n_sites` +1.22, `verified` +1.03, `mean_sup` +0.90, `min_sup` -0.88, `hand_conf` +0.86, `tspan` -0.85.

Dump _qf1 — 9,566 training candidates (seeds 500–502), 16,454 evaluation candidates (seeds 0–4), positive rate 1.0%. Pooled AUC gold-vs-spurious: hand-set confidence 0.672, learned 0.831.

architecture	gold patterns	with any candidate	found (current ranker)	found (learned rank, same #kept)	found (learned, p≥0.5)	AUC current	AUC learned
H_mycelic_full	200	159	0.300	0.560	0.545	0.672	0.831

Largest standardised weights: `mean_lag` -1.41, `dispersion` -1.24, `attribution` -1.12, `verified` +0.99, `q_evidence_frac` -0.91, `tspan` -0.81, `n_links` +0.78.

9. vNext, measured so far

Everything in this section is a **paired** experiment on the evaluation seeds (0–4 at 10,000 users): the variant and its base run on identical worlds, and the Δ shown under each variant value is the paired mean with an arrow when its 95% bootstrap interval excludes zero. Every knob in every variant was fitted on the calibration seeds (500–502) and frozen before these seeds were touched. Nothing here changes the register, the threshold, the gold labels or the matching rule.

9.1 A ranker fitted on kernel-side features

The confidence logistic is replaced as a *ranker* only. Version 1 kept the hand-set ≥ 0.5 gate and re-ordered inside it; version 2 keeps the hand-gated **count** and lets the learned score choose which candidates fill those slots, with anchor-level context features (how many candidates share the entity, where this one ranks among them). Fitted once on the pooled candidates of all four systems on seeds 500–502; l2 and the interaction set chosen leave-one-seed-out on those seeds only.

Version 1 — order within the hand gate (base = same system, hand-set ranker)

experiment	seeds	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
hierarchy, calibrated budget: hand ranker	5	0.375	0.215	0.690	0.0227	0.975	0.180	1.09e+06	9.10e+04
hierarchy, calibrated budget: variant	5	0.440 ^{+0.065 ↑}	0.167 ^{-0.048 ↓}	0.685 ^{-0.005 ↓}	0.1318 ^{+0.1091 ↑}	0.970 ^{-0.004 ↓}	0.245 ^{+0.065 ↑}	1.08e+06 ^{-6.10e+03 ↓}	9.08e+04 ^{-2.47e+02 ↓}
hierarchy, full question budget: hand ranker	5	0.375	0.215	0.690	0.0227	0.975	0.180	1.09e+06	9.10e+04
hierarchy, full question budget: variant	5	0.450 ^{+0.075 ↑}	0.197 ^{-0.018 ↓}	0.980 ^{+0.290 ↑}	0.0573 ^{+0.0346 ↑}	0.970 ^{-0.005 ↓}	0.240 ^{+0.060 ↑}	2.17e+06 ^{+1.08e+06 ↑}	2.26e+05 ^{+1.35e+05 ↑}
A2_chunked_ctx: hand ranker	5	0.685	0.653	1.000	0.0669	0.954	0.345	2.08e+06	3.00e+00
A2_chunked_ctx: variant	5	0.650 ^{-0.035 ↓}	0.561 ^{-0.092 ↓}	1.000 ^{+0.000}	0.1831 ^{+0.1161 ↑}	0.957 ^{+0.003 ↑}	0.350 ^{+0.005}	2.08e+06 ^{+0.00e+00}	3.00e+00 ^{+0.00e+00}
B4_central_triage: hand ranker	5	0.370	0.258	0.970	0.0219	0.975	0.175	3.98e+05	1.00e+04
B4_central_triage: variant	5	0.505 ^{+0.135 ↑}	0.238 ^{-0.020 ↓}	0.970 ^{+0.000}	0.1192 ^{+0.0972 ↑}	0.966 ^{-0.009 ↓}	0.240 ^{+0.065 ↑}	3.98e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}
Y_oracle_retrieval: hand ranker	5	0.240	0.143	1.000	0.0061	0.984	0.125	4.87e+05	1.00e+04
Y_oracle_retrieval: variant	5	0.360 ^{+0.120 ↑}	0.120 ^{-0.023 ↓}	1.000 ^{+0.000}	0.0232 ^{+0.0171 ↑}	0.976 ^{-0.008 ↓}	0.200 ^{+0.075 ↑}	4.87e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}

Version 2 — learned top-K gate + anchor context

experiment	seeds	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
hierarchy, calibrated budget: hand ranker	5	0.375	0.215	0.690	0.0227	0.975	0.180	1.09e+06	9.10e+04
hierarchy, calibrated budget: variant	5	0.480 ^{+0.105 ↑}	0.253 ^{+0.038 ↑}	0.690 ^{+0.000}	0.1334 ^{+0.1107 ↑}	0.968 ^{-0.007 ↓}	0.275 ^{+0.095 ↑}	1.08e+06 ^{-1.17e+04 ↓}	9.07e+04 ^{-3.22e+02 ↓}
A2_chunked_ctx: hand ranker	5	0.685	0.653	1.000	0.0669	0.954	0.345	2.08e+06	3.00e+00
A2_chunked_ctx: variant	5	0.645 ^{-0.040 ↓}	0.566 ^{-0.087 ↓}	1.000 ^{+0.000}	0.1999 ^{+0.1329 ↑}	0.957 ^{+0.003 ↑}	0.380 ^{+0.035}	2.08e+06 ^{+0.00e+00}	3.00e+00 ^{+0.00e+00}
B4_central_triage: hand ranker	5	0.370	0.258	0.970	0.0219	0.975	0.175	3.98e+05	1.00e+04
B4_central_triage: variant	5	0.540 ^{+0.170 ↑}	0.280 ^{+0.022 ↑}	0.970 ^{+0.000}	0.1298 ^{+0.1078 ↑}	0.964 ^{-0.011 ↓}	0.295 ^{+0.120 ↑}	3.98e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}
Y_oracle_retrieval: hand ranker	5	0.240	0.143	1.000	0.0061	0.984	0.125	4.87e+05	1.00e+04
Y_oracle_retrieval: variant	5	0.360 ^{+0.120 ↑}	0.104 ^{-0.040 ↓}	1.000 ^{+0.000}	0.0210 ^{+0.0150 ↑}	0.976 ^{-0.008 ↓}	0.230 ^{+0.105 ↑}	4.87e+05 ^{+0.00e+00}	1.00e+04 ^{+0.00e+00}

What it says. The same ranker lifts the hierarchy, its centralised twin and the perfect-retrieval oracle by 10–17 points of discovery and multiplies AP by four to six, and it makes `A2_chunked_ctx` *worse* on both found and rare recall. A ranker chosen on one system's candidates must not be imposed on another, so adoption is now decided per architecture on the calibration seeds — each system keeps whichever ranker is not worse there — and stored with the other calibrated settings. Rare recall on the hierarchy moves inside noise in both versions: the ranker recovers common patterns first.

Two regressions, not hidden. Decoy acceptance rises with the ranker on every system that adopts it (hierarchy 0.180 → 0.275, B4 0.175 → 0.295, oracle 0.125 → 0.230): the planted traps share the features that make a genuine chain look genuine, and the ranker promotes them alongside. FDR falls slightly at the same time, so the register is cleaner overall but more of what it contains is a trap rather than noise. This is the next thing the ranker has to be taught, and it is the reason decoy acceptance stays a headline metric rather than a footnote.

9.2 Spending the question budget, with the ranker on

Base = the hierarchy with ranker v2 at its calibrated budget (question_frac = 0.25). The earlier sweep without the ranker turned over at 0.35 and *lost* discovery at 1.0; the question is whether a better ranker lets the recovered evidence be reported.

experiment	seeds	found	rare recall	evidence cov.	AP	FDR	decoy acc.	compute	calls
question_frac 0.50: qf 0.25	5	0.480	0.253	0.690	0.1334	0.968	0.275	1.08e+06	9.07e+04
question_frac 0.50: variant	5	0.510 +0.030	0.258 +0.005	0.875 +0.185 ↑	0.1447 +0.0114	0.966 -0.002	0.300 +0.025	1.41e+06 +3.38e+05 ↑	1.33e+05 +4.28e+04 ↑
question_frac 0.65: qf 0.25	5	0.480	0.253	0.690	0.1334	0.968	0.275	1.08e+06	9.07e+04
question_frac 0.65: variant	5	0.520 +0.040	0.318 +0.065	0.970 +0.280 ↑	0.1336 +0.0002	0.965 -0.003	0.300 +0.025	1.63e+06 +5.55e+05 ↑	1.61e+05 +7.05e+04 ↑
question_frac 0.80: qf 0.25	5	0.480	0.253	0.690	0.1334	0.968	0.275	1.08e+06	9.07e+04
question_frac 0.80: variant	5	0.510 +0.030	0.318 +0.065	0.985 +0.295 ↑	0.1191 -0.0143	0.966 -0.002	0.285 +0.010	1.87e+06 +7.98e+05 ↑	1.91e+05 +1.00e+05 ↑
question_frac 1.00: qf 0.25	5	0.480	0.253	0.690	0.1334	0.968	0.275	1.08e+06	9.07e+04
question_frac 1.00: variant	5	0.515 +0.035 ↑	0.330 +0.077	0.985 +0.295 ↑	0.0974 -0.0360 ↓	0.966 -0.002 ↓	0.280 +0.005	2.20e+06 +1.13e+06 ↑	2.31e+05 +1.40e+05 ↑

What the sweep says. With the ranker on, every budget above 0.25 recovers evidence (coverage 0.69 → 0.87–0.985) and reports a little more of it (found +0.03 to +0.04, rare +0.065 to +0.077 — the rare gain is where the extra evidence goes), at +31% to +104% compute. The budget no longer *hurts* discovery as it did without the ranker, but it is not converting either: at question_frac 1.0 the kernel holds evidence for 98.5% of patterns and reports 51.5%, against a measured ranking ceiling of 79.5%. The remaining loss is inside the kernel's ordering of candidates it already has, which is where the next iteration goes.

9.3 Where a full budget spends its compute, and what batching returns

Profiled on one calibration seed (500) at 10,000 users with the full question budget, single runs, **not** evaluation-seed measurements:

stage	share of compute	calls	note
kernel re-read of the merged pool	42%	1	89,525 objects × 26 tokens in one frontier call
descent routing	31%	145,823	one frontier-tier call per (parent node, anchor)
user reads	6%	69,549	8,361 of 9,491 reached users queried more than once
edge extraction	6%	9,999	unchanged by any of this

Three cost changes, same seed, full budget, ranker off:

change	found	AP	rare	pool	compute	calls	kept?
none	0.400	0.0123	0.125	89,525	2.31e+06	227,524	—
batched routing + reads (one call per node / per user; identical decisions and per-record tokens)	0.400	0.0123	0.125	89,525	1.91e+06	27,127	yes — pure cost
+ merge descent returns per (predicate, entity) before the kernel reads them	0.275	0.0052	—	22,225	1.07e+06	27,128	no
+ merge per (predicate, entity, polarity)	0.325	0.0075	0.312	23,210	1.08e+06	27,128	not yet — needs the ranker re-fitted on merged candidates

9.4 Targeted local re-extraction

When a descent reaches a user, that user re-reads its *own* notes that name the entity but produced no claim on the first pass (the edge model's recall is a per-record coin; §5 measured that a fresh read recovers 80–91% of the patterns lost there). Raw text stays on the node; only objects leave. One calibration seed, full budget, ranker v2, batched:

	found	AP	rare	evidence cov.	compute	records re-read locally
without	0.350	0.0818	0.062	0.925	1.97e+06	0
with	0.475	0.0385	0.188	0.925	1.99e+06	7,980

The paired evaluation-seed test of this is queued behind the budget sweep; it is reported here as a single-seed signal, not a result.

9.5 What is accepted so far, and at what cost

change	status	Δ found (10k, paired, 5 seeds)	Δ compute
ranker v2, per-architecture adoption	accepted	+0.105 on the hierarchy	−1%
batched descent metering	accepted (cost only)	0	−17% at full budget, −88% calls
question budget above 0.25	under test	see 9.2	+31% at 0.50
kernel-side evidence merge	rejected for now	−0.075 (one cal seed)	−44%
local re-extraction	promising, untested on eval seeds	+0.125 (one cal seed)	+0.7%
support-1 sketch bits for rare facets	not started	—	—

7. What this changes about the next architecture

Not "richer sketches". In order of the loss they address:

1. **A calibrated ranker** over kernel-side features, fitted on held-out seeds by the same protocol as every other knob, so the full question budget can be spent without flooding the register. Targets the register loss (12% at 10k; binding at 50k once the budget is raised).
2. **A cheaper descent**, so the full budget is affordable at 50k (it currently doubles compute and calls). Targets the question-budget loss.
3. **Targeted local re-extraction** on descent. Targets the extraction loss.
4. **A support-1 sketch bit** for rare facets, *if* (1) absorbs the extra triage noise. Targets the sketch loss at scale.

Each is a paired experiment against `H_mycelic_full` on identical worlds, with the centralised controls re-run at the same time. None moves raw text.

8. Reproducing this

```
python3 -m research.mycelic.loss_accounting      # 10k x 5, 50k x 3 -> loss_funnel.jsonl
python3 -m research.mycelic.loss_report          # the tables
python3 -m research.mycelic.quick_paired --tag diag_qf1 --cfg '{"question_frac": 1.0}' --seeds 0,1,2,3,4
python3 -m research.mycelic.calibrator dump && python3 -m research.mycelic.calibrator fit
python3 -m research.mycelic.loss_doc             # this document
python3 -m research.mycelic.make_pdf docs/MYCELIC_LOSS_ACCOUNTING.md docs/MYCELIC_LOSS_ACCOUNTING.pdf
```

Generated 2026-09-22 by `research/mycelic/loss_doc.py` from the raw artifacts.